Human Grounding in Increasingly Capable AI Systems

A Decision-Information Theory of External Grounding, Human–AI Complementarity, and Constructive Reference Formation

Working paper draft v0.9

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Status. Version v0.9 extends the v0.8 one-shot and bounded-query theory into causal and controlled settings. For exogenous sequential processes, ideal grounding is identified with a nonanticipative decision rate–distortion functional and is shown to depend only on the causal decision-profile process. For finite controlled Markov systems, exact dynamic decision quotients preserve the complete causal information–distortion frontier. Approximate quotients now receive explicit finite-horizon distortion guarantees and, for finite action alphabets, an explicit directed-information continuity bound yielding a two-sided approximate frontier theorem. The draft also incorporates the first HG-v2 implementation results: routing and query synthesis are empirically separable, synthesized queries reduce external response cost on a frozen 3B model, and the experiments expose capability-floor, soundness/minimality, and measurement-accounting requirements that motivate concrete training and architecture changes. This draft has not received independent external mathematical review.

Purpose of this version. This draft is written for mathematical auditability. Every object is introduced with a type or function signature before it is used. All mathematical notation, including inline object names, is typeset in LaTeX math mode. Each theorem states the assumptions under which it is valid and distinguishes exact equality from lower bounds, structural hypotheses, and model-specific corollaries.

0. Reader's guide

The central question is not whether a sufficiently capable AI must literally keep a person clicking "approve." The question is whether, after the AI has optimally used everything already inside the system, a positive amount of *decision-relevant* information must still cross the system boundary from the external world or from a protected human/reference process.

The key distinction is between the full reference state Z^{ref} and the part of that state that changes the action. A system need not reconstruct the entire world or human. It needs only enough information to act within a specified distortion tolerance D.

Four mechanisms are kept separate throughout:

- **Information requirement:** how many fresh decision-relevant bits are missing after conditioning on the complete internal state M.
- **Protocol overhead:** how much additional communication is required because the actual source, such as a human, does not know exactly what the AI already knows or what distinction it needs resolved.
- **Consequence:** how costly it is to act without obtaining the missing information.
- **Integrity:** whether the reference becomes predictable through legitimate learning and deliberation or through disallowed target steering.

0.1 Minimal information-theory toolkit

For a discrete random variable X, Shannon entropy is

$$H(X) = -\sum_{x} p(x) \log_2 p(x).$$

Conditional entropy $H(X \mid M)$ is uncertainty remaining in X after M is known. Conditional mutual information is

$$I(X;Y \mid M) = H(X \mid M) - H(X \mid M,Y).$$

If A is generated from (M, Y), then conditional data processing gives

$$I(Z; A \mid M) \le I(Z; Y \mid M).$$

Rate–distortion theory asks for the minimum mutual information needed to achieve an expected loss no larger than D. In this paper the reproduction symbol is usually an action A, so the central rate–distortion object is decision-relative.

0.2 Status labels

Label	Meaning
Definition	Introduces an object or relation.
Theorem / Lemma	Proved inside the explicitly stated mathematical model.
Corollary	Follows from a theorem plus additional stated assumptions.
Model-specific result	Exact only for the stated binary, Gaussian, or finite construction.
Structural hypothesis	A property real capability sequences may or may not satisfy.
Empirical result	Observed in the existing experiments; not a theorem.
Open problem	Not established here.

1 Problem statement and primitive objects

The literal statement that an AGI must always benefit from explicit human supervision is false. Fully observed deterministic systems, objectives unrelated to humans, already-resolved decision problems, and perfectly predictable reference processes are direct counterexamples.

The defensible question is:

> For human-relative open decision problems, after optimal use of the agent's complete internal information, what decision-relevant uncertainty remains; how much fresh external information is required to meet a target distortion; how does that requirement scale with capability; and which part, if any, must originate from a protected human/reference process?

1.1 Primitive spaces and random variables

For each capability level C, let:

- $\mathcal{Z}_C$ be a reference-state space;
- $Z_C^{\mathrm{ref}} : \Omega \to \mathcal{Z}_C$ be the protected reference-relevant random state;
- $\mathcal{M}_C$ be an internal-information state space;
- $M_C : \Omega \to \mathcal{M}_C$ be the AI's complete usable information before new grounding arrives;
- $\mathcal{A}_C$ be the available action or policy set;
- $A_C : \Omega \to \mathcal{A}_C$ be the final action;
- $d_C : \mathcal{Z}_C \times \mathcal{A}_C \to [0, \infty)$ be a decision distortion;
- $D \in [0, \infty)$ be the tolerated expected distortion.

When utility is more natural than distortion, define

$$u_C : \mathcal{A}_C \times \mathcal{Z}_C \to \mathbb{R}$$

and

$$u_C^*(z) = \sup_{a \in \mathcal{A}_C} u_C(a, z),$$

then use regret distortion

$$d_C(z, a) = u_C^*(z) - u_C(a, z).$$

1.2 The anchor

The random variable Z_C^{ref} is not automatically the currently observed human state. It is defined relative to an externally specified protected causal or normative reference process. Legitimate teaching, explanation, persuasion, or deliberation may change the human, so the anchor need not be a no-influence counterfactual. The theory requires the allowed causal pathways to be specified outside the information objective.

2 Ideal decision grounding

2.1 Message grounding

Let $Y : \Omega \to \mathcal{Y}$ be a fresh external message. In the unrestricted ideal model, an encoder may use both Z_C^{ref} and M_C:

$$P(Y \mid Z_C^{\mathrm{ref}}, M_C).$$

The decoder/action rule is

$$P(A_C \mid M_C, Y).$$

Definition 2.1 (Ideal message grounding)**.** For fixed C, M_C, and D, define

$$G_{\mathrm{msg}}(C, M_C, D) = \inf I(Z_C^{\mathrm{ref}}; Y \mid M_C),$$

where the infimum is over all message channels and decoders satisfying

$$\mathbb{E}\big[d_C(Z_C^{\mathrm{ref}}, A_C)\big] \le D.$$

2.2 Decision rate–distortion

Definition 2.2 (Conditional decision rate–distortion)**.** Define

$$R_{\mathrm{dec}}(C, M_C, D) = \inf_{P(A_C \mid Z_C^{\mathrm{ref}}, M_C)} I(Z_C^{\mathrm{ref}}; A_C \mid M_C),$$

subject to

$$\mathbb{E}\big[d_C(Z_C^{\mathrm{ref}}, A_C)\big] \le D.$$

The quantity $R_{\mathrm{dec}}(C, M_C, D)$ is the fundamental information deficit after the complete internal state M_C is accounted for. It does not count interaction turns or raw message length. It counts the minimum fresh decision-relevant mutual information.

Theorem 2.3 (Ideal grounding identity)**.** *Under the unrestricted ideal message model,*

$$G_{\mathrm{msg}}(C, M_C, D) = R_{\mathrm{dec}}(C, M_C, D).$$

Proof. For any feasible external message Y, the final action A_C is produced from (M_C, Y). Conditional data processing gives

$$I(Z_C^{\mathrm{ref}}; A_C \mid M_C) \le I(Z_C^{\mathrm{ref}}; Y \mid M_C).$$

Thus every feasible message protocol costs at least $R_{\mathrm{dec}}(C, M_C, D)$, so

$$G_{\mathrm{msg}}(C, M_C, D) \ge R_{\mathrm{dec}}(C, M_C, D).$$

Conversely, for any feasible action channel $P(A_C \mid Z_C^{\mathrm{ref}}, M_C)$ choose the message $Y = A_C$. The decoder outputs A_C directly, and

$$I(Z_C^{\mathrm{ref}}; Y \mid M_C) = I(Z_C^{\mathrm{ref}}; A_C \mid M_C).$$

Taking infima proves equality. □

Remark 2.4. The converse is deliberately optimistic: the ideal encoder knows both Z_C^{ref} and M_C. A human, sensor, or one-way source usually does not have that information. Therefore $R_{\mathrm{dec}}(C, M_C, D)$ is a lower envelope. Actual acquisition cost may be larger.

Remark 2.5 (Encoder-side information is load-bearing). If the encoder sees only Z_C^{ref} and not M_C, equality in Theorem 2.3 can fail by an unbounded factor. The index construction in Appendix D has $R_{\mathrm{dec}} = H(J \mid M) = 1$ bit while a fixed-length zero-error one-way source ignorant of M must distinguish all 2^n source strings and can require n response bits. Thus $G_{\mathfrak{Q}} \ge R_{\mathrm{dec}}$ is the correct general statement for restricted source knowledge.

2.3 Restricted external channels

Let $\mathfrak{Q}$ be a class of realizable grounding protocols. Define

$$G_{\mathfrak{Q}}(C, M_C, D)$$

by restricting the infimum in G_{msg} to $\mathfrak{Q}$. Then

$$G_{\mathfrak{Q}}(C, M_C, D) \geq R_{\text{dec}}(C, M_C, D).$$

The gap is protocol inefficiency, communication asymmetry, or physical acquisition cost rather than fundamental missing decision information.

3 Internal information and exact decisions

3.1 Closed computation cannot create fresh external information

Let R be internally generated randomness and let

$$W = f(M_C, R), \qquad f : \mathcal{M}_C \times \mathcal{E} \to \mathcal{W}.$$

Assume

$$R \perp\!\!\!\perp Z_C^{\text{ref}} \mid M_C.$$

Then

$$I(Z_C^{\text{ref}}; W \mid M_C) = 0.$$

This does not say additional reasoning cannot improve a current prediction. It says that if M_C already denotes the complete usable internal information state, a variable generated only from M_C cannot add information about an external reference state beyond M_C.

3.2 Internal-information monotonicity

Theorem 3.1 (More nested internal information cannot increase ideal grounding)**.** *Suppose there exists a deterministic map*

$$g : \mathcal{M}_2 \to \mathcal{M}_1$$

such that

$$M_1 = g(M_2).$$

For a fixed decision problem C and target D,

$$R_{\text{dec}}(C, M_2, D) \leq R_{\text{dec}}(C, M_1, D).$$

Proof. Any decision channel feasible using M_1 can be reproduced by a system with M_2 by computing $M_1 = g(M_2)$ and ignoring the extra information. For that reproduced channel, the conditional chain rule implies

$$I(Z^{\text{ref}}; A \mid M_2) \leq I(Z^{\text{ref}}; A \mid M_1).$$

Taking infima over feasible channels gives the result. □

Remark 3.2 (Why nesting matters). A larger entropy or even a larger mutual information $I(Z^{\text{ref}}; M)$ is not a substitute for the structural relation $M_1 = g(M_2)$. Appendix D gives a finite counterexample in which M_2 contains more information about Z^{ref} in aggregate but omits the decision-relevant component known in M_1, causing grounding to increase. The theorem also extends from deterministic nesting to Markov degradation $Z^{\text{ref}} \perp\!\!\!\perp M_1 \mid M_2$.

3.3 Decision-profile sufficiency

The full protected state can contain information that never changes the loss of any available action. The following theorem shows that such information is exactly removable from the ideal decision-rate problem.

Let $T = t(Z)$ for a deterministic map

$$t : \mathcal{Z} \to \mathcal{T},$$

and suppose there exists

$$\widetilde{d} : \mathcal{T} \times \mathcal{A} \to [0, \infty)$$

such that

$$d(z, a) = \widetilde{d}(t(z), a)$$

for every (z, a). Define

$$R_Z(D \mid M) = \inf_{P(A|Z,M)} I(Z; A \mid M),$$

subject to $\mathbb{E}[d(Z, A)] \le D$, and

$$R_T(D \mid M) = \inf_{P(A|T,M)} I(T; A \mid M),$$

subject to $\mathbb{E}[\widetilde{d}(T, A)] \le D$.

Theorem 3.3 (Decision-profile sufficiency)**.** *Under the finite-alphabet assumptions above, for every feasible distortion level D,*

$$\boxed{R_Z(D \mid M) = R_T(D \mid M).}$$

Proof. Take first any feasible reduced channel $P(A \mid T, M)$ and lift it to the full state by setting

$$P(A \mid Z = z, M = m) = P(A \mid T = t(z), M = m).$$

The distortion is unchanged because $d(Z, A) = \widetilde{d}(T, A)$. Since T is a deterministic function of Z and the lifted channel obeys $A \perp\!\!\!\perp Z \mid (T, M)$,

$$I(Z; A \mid M) = I(T, Z; A \mid M) = I(T; A \mid M).$$

Hence $R_Z(D \mid M) \le R_T(D \mid M)$.

Conversely, take any feasible full-state channel $P(A \mid Z, M)$ and define its Markovized version

$$\overline{P}(a \mid t, m) = P(A = a \mid T = t, M = m) = \sum_z P(a \mid z, m) P(z \mid t, m).$$

This preserves the joint marginal law of (T, M, A) and therefore preserves expected distortion. The resulting reduced-channel cost is

$$I_{\overline{P}}(T; A \mid M) = I_P(T; A \mid M) \le I_P(Z; A \mid M),$$

where the inequality follows from conditional data processing because $T = t(Z)$. Thus every feasible full-state channel has a feasible reduced replacement of no greater information cost, giving $R_T(D \mid M) \le R_Z(D \mid M)$. Combining the two inequalities proves equality. □

A canonical choice is the regret profile

$$T_C(z) = \big(d_C(z, a)\big)_{a \in \mathcal{A}_C}.$$

Equivalently, define

$$z \sim_C z' \quad \Longleftrightarrow \quad d_C(z, a) = d_C(z', a) \text{ for every } a \in \mathcal{A}_C.$$

The quotient $\mathcal{Z}_C / \sim_C$ is the coarsest deterministic decision profile, up to relabeling, that preserves the entire distortion geometry. Theorem 3.3 therefore gives a precise meaning to *decision-relevant information*: ideal grounding depends on the protected state only through this quotient whenever the stated factorization holds.

Corollary 3.4 (Canonical decision quotient)**.** *Let S_C^* denote the equivalence class of Z_C^{ref} under $\sim_C$. Then*

$$R_{\mathrm{dec}}(C, M, D) = R_{S_C^*}(D \mid M).$$

At $D = 0$ with unique optimal actions, the still coarser statistic J_C suffices for exact action identification and Theorem 3.5 below gives the corresponding entropy identity.

3.4 Unique zero-regret decisions

Assume finite $\mathcal{Z}_C$ and $\mathcal{A}_C$, and suppose every state has a unique optimal action

$$j_C : \mathcal{Z}_C \to \mathcal{A}_C, \qquad j_C(z) = \arg\max_{a \in \mathcal{A}_C} u_C(a, z).$$

Define the correct-decision random variable

$$J_C = j_C(Z_C^{\mathrm{ref}}).$$

Theorem 3.5 (Exact decision grounding)**.** *Under unique optimal actions and zero expected regret,*

$$R_{\mathrm{dec}}(C, M_C, 0) = H(J_C \mid M_C).$$

Proof. Zero expected nonnegative regret implies zero regret almost surely. Uniqueness forces $A_C = J_C$ almost surely, hence

$$H(J_C \mid M_C, A_C) = 0.$$

Because J_C is a deterministic function of Z_C^{ref},

$$I(Z_C^{\mathrm{ref}}; A_C \mid M_C) \geq I(J_C; A_C \mid M_C) = H(J_C \mid M_C).$$

Choosing $A_C = J_C$ achieves equality. □

3.5 Non-unique optima

If

$$\mathcal{O}_C(z) = \{a \in \mathcal{A}_C : d_C(z, a) = 0\}$$

contains more than one action, arbitrary tie-breaking can overstate grounding. The exact zero-regret object is

$$\inf I(Z_C^{\mathrm{ref}}; A_C \mid M_C)$$

subject to

$$\Pr\big(A_C \in \mathcal{O}_C(Z_C^{\mathrm{ref}})\big) = 1.$$

A subset $S \subseteq \mathcal{Z}_C$ may share a common zero-regret message cell only when

$$\bigcap_{z \in S} \mathcal{O}_C(z) \neq \varnothing.$$

This connects the finite zero-error case to functional source coding and graph/hypergraph entropy. Appendix D gives an enumeration-verified two-state example in which tie-breaking claims one bit while a common zero-regret action requires zero bits. We do not claim novelty for the underlying functional-coding machinery.

4 Capability and the information race

4.1 Decision refinement

Capability does not automatically make grounding harder. A new action may dominate all old actions and collapse the decision problem. Appendix D records this as an explicit assumption knockout. Therefore the following structural relation, rather than action-set inclusion alone, is what yields monotonicity.

Definition 4.1 (Distortion-preserving decision refinement)**.** Capability C_2 refines C_1 if there exists

$$\phi : \mathcal{A}_{C_2} \to \mathcal{A}_{C_1}$$

such that for all z and a_2,

$$d_{C_1}\big(z, \phi(a_2)\big) \leq d_{C_2}(z, a_2).$$

Theorem 4.2 (Capability-refinement monotonicity)**.** *If C_2 refines C_1, then for every fixed internal state M and target D,*

$$R_{\mathrm{dec}}(C_2, M, D) \geq R_{\mathrm{dec}}(C_1, M, D).$$

Proof. For any C_2-feasible action A_2, define $A_1 = \phi(A_2)$. The pointwise refinement condition gives

$$\mathbb{E}[d_{C_1}(Z^{\mathrm{ref}}, A_1)] \le \mathbb{E}[d_{C_2}(Z^{\mathrm{ref}}, A_2)] \le D.$$

By data processing,

$$I(Z^{\mathrm{ref}}; A_1 \mid M) \le I(Z^{\mathrm{ref}}; A_2 \mid M).$$

Taking infima proves the result. □

4.2 General information-race decomposition

Compare (C_1, M_1) with (C_2, M_2) and insert the counterfactual pair (C_2, M_1). Define capability pressure

$$F_{1\to 2}(D) = R_{\mathrm{dec}}(C_2, M_1, D) - R_{\mathrm{dec}}(C_1, M_1, D),$$

and internal-learning relief

$$L_{1\to 2}(D) = R_{\mathrm{dec}}(C_2, M_1, D) - R_{\mathrm{dec}}(C_2, M_2, D).$$

Then exactly

$$R_{\mathrm{dec}}(C_2, M_2, D) - R_{\mathrm{dec}}(C_1, M_1, D) = F_{1\to 2}(D) - L_{1\to 2}(D).$$

This identity is algebraic. Under Theorem 4.2, $F_{1\to 2}(D) \ge 0$. Under Theorem 3.1, $L_{1\to 2}(D) \ge 0$.

4.3 Nested exact-decision refinement

If

$$J_1 = f(J_2)$$

and

$$M_1 = g(M_2),$$

then Theorem 3.5 and the entropy chain rule give

$$G_2(0) - G_1(0) = H(J_2 \mid J_1, M_2) - \big[H(J_1 \mid M_1) - H(J_1 \mid M_2)\big].$$

The first term is unresolved new decision refinement; the second is old decision uncertainty removed by the richer internal state.

5 The master grounding theorem

We now state the common lower bound that contains the static, dynamic, human-complementarity, and constructive-reference cases as specializations.

5.1 Causal local setup

At time t, let:

- $Z_t^{\mathrm{ref},\pi}$ be the protected reference state induced under an allowed interaction/control policy π;
- M_t be the complete AI information state immediately before new grounding;
- $Q_t = q_t(M_t)$ be a query or intervention chosen from M_t;
- Y_t be fresh external information arriving after Q_t;
- A_t be generated from (M_t, Q_t, Y_t).

The action generation condition is

$$A_t \perp\!\!\!\perp Z_t^{\mathrm{ref},\pi} \mid (M_t, Q_t, Y_t).$$

Define local decision rate–distortion

$$R_{C,t}^{\pi}(D) = \inf_{P(A \mid Z_t^{\mathrm{ref},\pi}, M_t, Q_t)} I(Z_t^{\mathrm{ref},\pi}; A \mid M_t, Q_t),$$

subject to

$$\mathbb{E}\big[d_{C,t}(Z_t^{\mathrm{ref},\pi}, A)\big] \le D.$$

Theorem 5.1 (Master grounding theorem)**.** *If an actual external message Y_t and decoder produce an action A_t with expected distortion*

$$D_t = \mathbb{E}\big[d_{C,t}(Z_t^{\mathrm{ref},\pi}, A_t)\big],$$

then

$$I(Z_t^{\mathrm{ref},\pi}; Y_t \mid M_t, Q_t) \geq R_{C,t}^{\pi}(D_t).$$

Consequently, defining cumulative local grounding by

$$G_{1:T}^{\pi} = \sum_{t=1}^{T} I(Z_t^{\mathrm{ref},\pi}; Y_t \mid M_t, Q_t),$$

we have

$$G_{1:T}^{\pi} \geq \sum_{t=1}^{T} R_{C,t}^{\pi}(D_t).$$

Proof. Because A_t is generated from (M_t, Q_t, Y_t), conditional data processing yields

$$I(Z_t^{\mathrm{ref},\pi}; A_t \mid M_t, Q_t) \leq I(Z_t^{\mathrm{ref},\pi}; Y_t \mid M_t, Q_t).$$

By the definition of $R_{C,t}^{\pi}(D_t)$,

$$I(Z_t^{\mathrm{ref},\pi}; A_t \mid M_t, Q_t) \geq R_{C,t}^{\pi}(D_t).$$

Combining the inequalities proves the local statement. Summing over t proves the cumulative lower bound. □

Remark 5.2 (Why this is a theorem schema rather than a global equality). In a feedback system, A_t or Q_t can change the distribution of future states $Z_{t+1}^{\mathrm{ref},\pi}$. Therefore separate minimization of each $R_{C,t}^{\pi}(D_t)$ need not solve the global sequential problem. The local lower bound is general; a full equality characterization requires causal or nonanticipative rate–distortion and directed-information machinery.

Remark 5.3 (Query measurability is part of the ledger). The condition $Q_t = q_t(M_t)$ is load-bearing. If a query can directly depend on $Z_t^{\mathrm{ref},\pi}$, then reference information can enter through Q_t while the displayed grounding term conditions it away. For example, $Q_t = Z_t^{\mathrm{ref},\pi}$, constant Y_t, and $A_t = Q_t$ can achieve zero distortion while $I(Z_t^{\mathrm{ref},\pi}; Y_t \mid M_t, Q_t) = 0$. If Q_t is not measurable from M_t, the ledger must charge the query information, e.g. $I(Z_t^{\mathrm{ref},\pi}; Q_t \mid M_t)$, as well.

6 Open systems and persistent innovation

6.1 IID innovations

Let $W_1, \ldots, W_T$ be IID with common law P_W. Let A_t incur distortion $d_C(W_t, A_t)$. Define

$$\mathcal{G}_{C,T}(D) = \frac{1}{T} \inf I(W^T; A^T),$$

subject to

$$\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[d_C(W_t, A_t)] \leq D.$$

Define the one-step rate–distortion function

$$R_C(D) = \inf_{P(A|W)} I(W; A)$$

subject to

$$\mathbb{E}[d_C(W, A)] \leq D.$$

Theorem 6.1 (IID innovation grounding)**.** *Under IID innovations and additive expected distortion,*

$$\mathcal{G}_{C,T}(D) = R_C(D)$$

for the ideal block model, up to standard infimum-attainment technicalities.

Proof. For the upper bound, apply an optimal or ε-optimal one-step channel independently at each time. Mutual information adds, giving total rate arbitrarily close to $TR_C(D)$.

For the lower bound, independence implies

$$I(W^T; A^T) \geq \sum_{t=1}^{T} I(W_t; A_t).$$

Let

$$D_t = \mathbb{E}[d_C(W_t, A_t)].$$

Then

$$I(W_t; A_t) \geq R_C(D_t).$$

Convexity of R_C and Jensen's inequality give

$$\frac{1}{T}\sum_{t=1}^{T} R_C(D_t) \geq R_C\left(\frac{1}{T}\sum_{t=1}^{T} D_t\right) \geq R_C(D).$$

Hence

$$I(W^T; A^T) \geq TR_C(D).$$

□

6.2 Blind-distortion threshold

Define

$$D_C^{\text{blind}} = \inf_{P(A)} \mathbb{E}[d_C(W, A)].$$

If $D \geq D_C^{\text{blind}}$, then $R_C(D) = 0$. Under ordinary nondegenerate rate–distortion conditions, if $D < D_C^{\text{blind}}$, then $R_C(D) > 0$.

Thus persistent randomness is not enough. The innovation must remain decision-relevant at the required tolerance.

6.3 Dependent processes: local view

For a dependent process, the local object remains

$$R_{C,t}(D) = \inf I(Z_t; A_t \mid M_t),$$

with the complete causal past already contained in M_t. Theorem 5.1 gives

$$I(Z_t; Y_t \mid M_t) \geq R_{C,t}(D_t).$$

This stagewise inequality is always a valid lower bound under the theorem's assumptions, but it is not by itself a general block equality because information acquired earlier can be retained and reused, actions can alter future states, and future reference states can depend on the interaction policy. The causal block formulation below makes those distinctions explicit.

7 Sequential and controlled grounding

7.1 Exogenous causal grounding and directed information

Let $Z^T = (Z_1, \ldots, Z_T)$ be an exogenous protected/reference process and let $X^T = (X_1, \ldots, X_T)$ denote causal non-grounding side information. At time t, only X^t is available. A nonanticipative ideal grounding encoder has a causal factorization of the form

$$P(Y^T \| Z^T, X^T) = \prod_{t=1}^{T} P(Y_t \mid Y^{t-1}, Z^t, X^t),$$

and a causal decision kernel has the form

$$P(A^T \| Y^T, X^T) = \prod_{t=1}^{T} P(A_t \mid A^{t-1}, Y^t, X^t).$$

Queries can be included in the causal history provided each query is measurable with respect to the pre-grounding state and therefore creates no uncharged path from the protected state.

Define causally conditioned directed information

$$I(Z^T \to Y^T \| X^T) := \sum_{t=1}^{T} I(Z^t; Y_t \mid Y^{t-1}, X^t),$$

and similarly

$$I(Z^T \to A^T \| X^T) := \sum_{t=1}^{T} I(Z^t; A_t \mid A^{t-1}, X^t).$$

For additive distortion

$$\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[d_t(Z_t, A_t)] \le D,$$

define the ideal causal message-grounding functional

$$G_T^{\mathrm{msg}}(D) := \inf I(Z^T \to Y^T \| X^T)$$

over nonanticipative grounding protocols, and define the nonanticipative decision rate–distortion functional

$$R_Z^{\mathrm{na},T}(D) := \inf I(Z^T \to A^T \| X^T)$$

over causal decision kernels satisfying the same distortion constraint.

Theorem 7.1 (Exogenous causal ideal-message equivalence)**.** *Assume the law of (Z^T, X^T) is exogenous to the action policy, grounding and decision kernels are nonanticipative, the ideal message channel is unrestricted, and all protected-state access is charged through the grounding channel. Then*

$$\boxed{G_T^{\mathrm{msg}}(D) = R_Z^{\mathrm{na},T}(D)}.$$

Proof. For any feasible grounding protocol, causal processing from Y^T to A^T gives the directed data-processing inequality

$$I(Z^T \to A^T \| X^T) \le I(Z^T \to Y^T \| X^T).$$

The induced action kernel is feasible for $R_Z^{\mathrm{na},T}(D)$, proving the converse. For achievability, take any feasible causal action kernel and choose the ideal grounding message to be the action itself, $Y_t = A_t$, with the decoder outputting $A_t = Y_t$. The distortion and directed information are then identical. Taking infima proves equality. □

The equality is an ideal information identity, not a claim that a practical human protocol literally communicates this many fixed-length bits. Query restrictions, one-way source ignorance, variable-length coding constraints, human computation, and other protocol restrictions can add overhead.

7.2 Causal decision-profile sufficiency

For each t, let

$$T_t = t_t(Z_t)$$

be a deterministic decision profile satisfying

$$d_t(z_t, a_t) = \widetilde{d}_t(t_t(z_t), a_t).$$

The canonical choice is

$$T_t(z_t) = \big(d_t(z_t, a)\big)_{a \in \mathcal{A}_t}.$$

Let $T^t = (T_1, \ldots, T_t)$.

Define

$$R_T^{\mathrm{na},T}(D) := \inf \sum_{t=1}^{T} I(T^t; A_t \mid A^{t-1}, X^t)$$

over causal kernels $P(A_t \mid A^{t-1}, T^t, X^t)$ satisfying the reduced distortion constraint.

Theorem 7.2 (Causal decision-profile sufficiency)**.** *Under the assumptions of Theorem 7.1, if the stage distortion factors through $T_t = t_t(Z_t)$, then*

$$\boxed{R_Z^{\mathrm{na},T}(D) = R_T^{\mathrm{na},T}(D)}.$$

Consequently,

$$\boxed{G_T^{\mathrm{msg}}(D) = R_T^{\mathrm{na},T}(D)}.$$

Proof. Any causal profile kernel can be lifted to the full state by substituting $T^t = t^t(Z^t)$. Under the lifted kernel,

$$A_t \perp\!\!\!\perp Z^t \mid (A^{t-1}, T^t, X^t),$$

so the chain rule gives

$$I(Z^t; A_t \mid A^{t-1}, X^t) = I(T^t; A_t \mid A^{t-1}, X^t).$$

Distortion is unchanged, proving

$$R_Z^{\mathrm{na},T}(D) \le R_T^{\mathrm{na},T}(D).$$

Conversely, start from an arbitrary feasible full-state causal kernel and define its causal Markovization by

$$\overline{P}(a_t \mid a^{t-1}, t^t, x^t) := P(A_t = a_t \mid A^{t-1} = a^{t-1}, T^t = t^t, X^t = x^t).$$

Because the source process is exogenous, induction shows that the new kernel preserves the joint law of (T^T, X^T, A^T). Hence it preserves distortion. Conditional data processing gives

$$I(T^t; A_t \mid A^{t-1}, X^t) \le I(Z^t; A_t \mid A^{t-1}, X^t)$$

under the original law, and preservation of the reduced joint law transfers the left-hand side to the Markovized process. Summing proves the reverse inequality. □

Corollary 7.3 (Unique zero-regret causal entropy)**.** *If each stage has a unique required action $J_t = j_t(Z_t)$ and zero distortion requires $A_t = J_t$, then under the same causal assumptions*

$$G_T^{\mathrm{msg}}(0) = \sum_{t=1}^{T} H(J_t \mid J^{t-1}, X^t) =: H(J^T \| X^T).$$

Thus information acquired at one stage is not charged again after it enters the causal state. For example, if all J_t equal a persistent hidden bit revealed at $t = 1$, only the first term need be positive. Ordinary block entropy $H(J^T \mid X^T)$ is generally not a substitute because it permits future observations to explain decisions that had to be made earlier.

7.3 Finite controlled Markov systems

The exogenous theorem does not cover an agent whose actions alter future states. Let $Z_t \in \mathcal{Z}$ be a finite controlled Markov state with transition kernel

$$P(z' \mid z, a)$$

and bounded stage distortion $d(z, a)$. Define the causal information–distortion frontier

$$R_Z^{\mathrm{ctrl},T}(D) := \inf_{\pi} \sum_{t=1}^{T} I(Z^t; A_t \mid A^{t-1})$$

over causal policies $\pi_t(a_t \mid z^t, a^{t-1})$ whose average distortion is at most D.

Let $\phi : \mathcal{Z} \to \mathcal{S}$ and write $S_t = \phi(Z_t)$. The quotient is an *exact dynamic decision quotient* if two conditions hold. First, distortion is preserved:

$$\phi(z) = \phi(z') \Longrightarrow d(z, a) = d(z', a) \quad \forall a.$$

Second, the controlled dynamics are lumpable: for every quotient class s', action a, and z, z' with $\phi(z) = \phi(z')$,

$$\sum_{\tilde{z}:\phi(\tilde{z})=s'} P(\tilde{z} \mid z, a) = \sum_{\tilde{z}:\phi(\tilde{z})=s'} P(\tilde{z} \mid z', a).$$

These conditions define an abstract distortion $\bar{d}(s, a)$ and abstract transition kernel $\bar{P}(s' \mid s, a)$.

Theorem 7.4 (Exact controlled decision-quotient sufficiency)**.** *For a finite controlled Markov system with an exact dynamic decision quotient $S_t = \phi(Z_t)$,*

$$\boxed{R_Z^{\mathrm{ctrl},T}(D) = R_S^{\mathrm{ctrl},T}(D)}$$

for every finite horizon T and feasible D.

Proof. Lift any quotient policy to the full process by replacing s^t with $\phi(z^t)$. Lumpability guarantees that the resulting (S^T, A^T) law is exactly the abstract law, while distortion preservation gives identical loss. The lifted policy satisfies

$$A_t \perp\!\!\!\perp Z^t \mid (S^t, A^{t-1}),$$

so its full-state and quotient directed-information terms coincide.

For the reverse direction, take any full-state policy and define

$$\bar{\pi}_t(a_t \mid s^t, a^{t-1}) = P_\pi(A_t = a_t \mid S^t = s^t, A^{t-1} = a^{t-1}).$$

Controlled lumpability implies by induction that this quotient policy preserves the entire joint law $P(S^T, A^T)$. Hence distortion is preserved, while

$$I(S^t; A_t \mid A^{t-1}) \le I(Z^t; A_t \mid A^{t-1})$$

by conditional data processing. Summation and optimization prove equality. □

This quotient is closely related to exact probabilistic bisimulation and MDP state abstraction. The point here is not to claim novelty for that equivalence relation, but to show that it preserves the entire grounding-relevant causal information–distortion frontier.

7.4 Approximate dynamic decision quotients

Exact lumpability is too strong for learned representations. Let $0 \le d(z, a), \bar{d}(s, a) \le d_{\max}$ and define the projected concrete transition

$$P_\phi(s' \mid z, a) = \sum_{z':\phi(z')=s'} P(z' \mid z, a).$$

Call $(\phi, \bar{d}, \bar{P})$ an $(\varepsilon_d, \varepsilon_p)$ approximate dynamic decision quotient if, whenever $s = \phi(z)$,

$$|d(z, a) - \bar{d}(s, a)| \le \varepsilon_d$$

and

$$\big\|P_\phi(\cdot \mid z, a) - \bar{P}(\cdot \mid s, a)\big\|_{\mathrm{TV}} \le \varepsilon_p.$$

Here $\|P - Q\|_{\mathrm{TV}} = \frac{1}{2} \sum_x |P(x) - Q(x)|$.

Define

$$\delta_t := \min\{1, (t-1)\varepsilon_p\}$$

and

$$\Delta_T^{\mathrm{abs}} := \varepsilon_d + \frac{d_{\max}}{T} \sum_{t=1}^{T} \delta_t.$$

Lemma 7.5 (Finite-horizon coupling bound)**.** *For any causal abstract policy and its concrete lift, there exists a coupling of the projected concrete and abstract histories such that*

$$\Pr\{(S^t, A^{t-1}) \neq (\bar{S}^t, \bar{A}^{t-1})\} \leq \delta_t.$$

Consequently,

$$\boxed{|D_T^{\text{conc}} - D_T^{\text{abs}}| \leq \Delta_T^{\text{abs}}}.$$

If $(T-1)\varepsilon_p \leq 1$*, this simplifies to*

$$\Delta_T^{\text{abs}} = \varepsilon_d + \frac{T-1}{2} d_{\max} \varepsilon_p.$$

Proof. Couple the initial quotient states identically. As long as the histories agree, the policies can be coupled to choose the same action. The two next-state distributions are within ε_p in total variation, so maximal coupling makes the next quotient states disagree with conditional probability at most ε_p. A union-bound induction gives the history mismatch bound δ_t. At stage t, direct distortion mismatch contributes at most ε_d, while history mismatch contributes at most $d_{\max}\delta_t$. Averaging proves the result. □

The same construction applied to the quotient Markovization of an arbitrary full-state policy shows that every full-state policy has a quotient-measurable concrete policy with average distortion at most $2\Delta_T^{\text{abs}}$ worse. Thus approximate dynamic quotients carry a finite-horizon decision-performance certificate, not merely a representation heuristic.

7.5 Directed-information continuity

To compare information frontiers rather than distortion alone, assume the action alphabet is finite with $m = |\mathcal{A}| \geq 2$. The following standard finite-alphabet continuity fact is the only additional ingredient.

Lemma 7.6 (Conditional-entropy continuity)**.** *Let* P_{UV} *and* Q_{UV} *be distributions on a finite alphabet with*

$$\|P_{UV} - Q_{UV}\|_{\text{TV}} \leq \delta.$$

If $|\mathcal{U}| = m$ *and* $0 \leq \delta \leq \frac{1}{2}$*, then*

$$|H_P(U \mid V) - H_Q(U \mid V)| \leq 2\delta \log_2 m + 2h_2(\delta).$$

For arbitrary δ*, the trivial bound* $\log_2 m$ *applies.*

Proof. This is the classical finite-alphabet conditional-entropy continuity inequality. A maximal coupling of (U, V) and (U', V') has mismatch probability at most δ. Conditioning on the mismatch indicator and applying the binary entropy bound to that indicator, together with $H(U) \leq \log_2 m$, yields the stated inequality. The bound is intentionally conservative; sharper constants do not affect the subsequent structural result. □

For $0 \leq \delta \leq 1$, define

$$g_m(\delta) := \begin{cases} \min\{\log_2 m, 3\delta \log_2 m + 2h_2(\delta)\}, & \delta \leq \frac{1}{2}, \\ \log_2 m, & \delta > \frac{1}{2}. \end{cases}$$

Lemma 7.7 (Same-kernel directed-information continuity)**.** *Consider two causal processes over* (S^T, A^T) *using the same action kernel*

$$K_t(a_t \mid s^t, a^{t-1})$$

but different history distributions. If

$$\left\|P(S^t, A^{t-1}) - Q(S^t, A^{t-1})\right\|_{\text{TV}} \leq \delta_t,$$

then

$$\left|I_P(S^t; A_t \mid A^{t-1}) - I_Q(S^t; A_t \mid A^{t-1})\right| \leq g_m(\delta_t).$$

Therefore

$$\boxed{\left|I_P(S^T \to A^T) - I_Q(S^T \to A^T)\right| \leq \eta_T}$$

with

$$\eta_T := \sum_{t=1}^{T} g_m(\delta_t).$$

Proof. Write

$$I(S^t; A_t \mid A^{t-1}) = H(A_t \mid A^{t-1}) - H(A_t \mid S^t, A^{t-1}).$$

The common action kernel implies

$$H(A_t \mid S^t, A^{t-1}) = \mathbb{E}\big[H\big(K_t(\cdot \mid S^t, A^{t-1})\big)\big].$$

The integrand lies in $[0, \log_2 m]$, so changing the history distribution by total variation at most δ_t changes this expectation by at most $\delta_t \log_2 m$. The induced joint distributions of (A_t, A^{t-1}) also differ by at most δ_t by data processing. Lemma 7.6 therefore bounds the first conditional-entropy difference by $2\delta_t \log_2 m + 2h_2(\delta_t)$ when $\delta_t \le \frac{1}{2}$. Adding the terms and using the trivial fact that a conditional mutual information with output alphabet $\mathcal{A}$ lies in $[0, \log_2 m]$ gives $g_m(\delta_t)$. Summation proves the directed-information bound. □

Theorem 7.8 (Approximate controlled information–distortion frontier)**.** *Let $R_Z^{\mathrm{ctrl},T}$ be the concrete full-state causal information–distortion frontier and $R_S^{\mathrm{ctrl},T}$ the abstract frontier of an $(\varepsilon_d, \varepsilon_p)$ approximate dynamic decision quotient. For finite action alphabet $|\mathcal{A}| = m$,*

$$\boxed{R_Z^{\mathrm{ctrl},T}\big(D + \Delta_T^{\mathrm{abs}}\big) \le R_S^{\mathrm{ctrl},T}(D) + \eta_T}$$

and

$$\boxed{R_S^{\mathrm{ctrl},T}\big(D + \Delta_T^{\mathrm{abs}}\big) \le R_Z^{\mathrm{ctrl},T}(D) + \eta_T}.$$

Proof. For the first inequality, take an ϵ-optimal abstract policy at distortion D and lift it to the concrete process. Lemma 7.5 increases distortion by at most Δ_T^{abs}. Because the lifted action depends on Z^t only through S^t, its concrete full-state directed information equals its concrete quotient directed information exactly. Lemma 7.7 compares that quotient information with the abstract-model directed information within η_T. Letting $\epsilon \downarrow 0$ proves the first inequality.

For the reverse inequality, start from an ϵ-optimal concrete full-state policy and define its quotient Markovization

$$\bar{\pi}_t(a_t \mid s^t, a^{t-1}) = P(A_t = a_t \mid S^t = s^t, A^{t-1} = a^{t-1}).$$

Convexity of total variation shows that the projected next-state kernel induced by the concrete policy remains within ε_p of the abstract kernel for each quotient history. The coupling argument therefore places the abstract distortion within Δ_T^{abs} of the concrete distortion. Under the concrete law,

$$I(S^t; A_t \mid A^{t-1}) \le I(Z^t; A_t \mid A^{t-1}),$$

and Lemma 7.7 changes the quotient information by at most η_T when transferring to the abstract law. Letting $\epsilon \downarrow 0$ proves the second inequality. □

The theorem deliberately compares frontiers after a distortion slack Δ_T^{abs}. Same-D closeness requires regularity of the rate–distortion curve in the distortion argument and is not claimed here. The explicit horizon dependence also matters: a representation can be excellent for short-horizon decisions and poor for long-horizon control even when its one-step transition error is small.

7.6 Capability floor and usable grounding

Empirical grounding curves can be confounded when a model cannot exploit information even when it is supplied perfectly. Let Π_C be the policy class implementable by a system of capability C, let S denote maximally available external information in the experiment, and define

$$D_C^{\mathrm{blind}} = \inf_{\pi \in \Pi_C} \mathbb{E}[d(Z, A)] \quad \text{using } M_C,$$

$$D_C^{\mathrm{full}} = \inf_{\pi \in \Pi_C} \mathbb{E}[d(Z, A)] \quad \text{using } (M_C, S),$$

and the unrestricted information oracle floor

$$D^{\mathrm{oracle}} = \inf_{P(A \mid M_C, S)} \mathbb{E}[d(Z, A)].$$

Then

$$D^{\mathrm{oracle}} \le D_C^{\mathrm{full}} \le D_C^{\mathrm{blind}}.$$

Define the model-constrained grounding requirement $G_C^{\Pi}(D)$ as the infimum external information over protocols whose final decoder lies in Π_C, with $\inf \varnothing = +\infty$.

Proposition 7.9 (Capability floor)**.**

$$D < D_C^{\text{full}} \Longrightarrow G_C^{\Pi}(D) = +\infty,$$

while

$$D \geq D_C^{\text{blind}} \Longrightarrow G_C^{\Pi}(D) = 0.$$

Thus the nontrivial empirically interpretable grounding region is

$$D_C^{\text{full}} \leq D < D_C^{\text{blind}}.$$

When $D_C^{\text{blind}} > D^{\text{oracle}}$, define the utilization ratio

$$U_C := \frac{D_C^{\text{blind}} - D_C^{\text{full}}}{D_C^{\text{blind}} - D^{\text{oracle}}} \in [0, 1].$$

This separates missing information from inability to use available information. A scaling study should not interpret communication efficiency below the capability floor as evidence about grounding demand.

8 Human–AI complementarity

8.1 Two-agent information structure

Let

$$M_A : \Omega \to \mathcal{M}_A$$

be the AI's private/internal information and

$$M_H : \Omega \to \mathcal{M}_H$$

be the human's private information. Let

$$J = f(Z^{\text{ref}})$$

be a unique required decision.

A one-way human message has the form

$$P(Y_H \mid M_H),$$

and the AI computes

$$\widehat{J} = \delta(M_A, Y_H).$$

8.2 Feasibility

Theorem 8.1 (Joint sufficiency for exact complementarity)**.** *Exact recovery by some one-way human-to-AI protocol is possible if and only if*

$$H(J \mid M_A, M_H) = 0.$$

Proof. If $H(J \mid M_A, M_H) > 0$, then even revealing all of M_H cannot make J deterministic given the AI state. Conversely, if $H(J \mid M_A, M_H) = 0$, the human can transmit $Y_H = M_H$, after which the AI can compute J. □

This separates three regimes:

$$H(J \mid M_A) = 0$$

means AI-only sufficiency;

$$H(J \mid M_A) > 0, \qquad H(J \mid M_A, M_H) = 0$$

means genuine complementarity; and

$$H(J \mid M_A, M_H) > 0$$

means the pair itself lacks sufficient information.

8.3 One-way communication lower bound

Define the exact one-way human information cost

$$C_{\to} = \inf I(M_H; Y_H \mid M_A),$$

where the infimum is over protocols satisfying

$$H(J \mid M_A, Y_H) = 0.$$

Theorem 8.2 (One-way human communication lower bound)**.** *In the jointly sufficient exact-decision regime,*

$$C_{\to} \geq H(J \mid M_A).$$

Proof. Exact recovery gives

$$I(J; Y_H \mid M_A) = H(J \mid M_A).$$

Because Y_H is generated from M_H, conditional data processing gives

$$I(J; Y_H \mid M_A) \leq I(M_H; Y_H \mid M_A).$$

Taking the infimum proves the bound. □

8.4 Coordination overhead

Definition 8.3 (Grounding coordination overhead)**.** Define

$$\Omega = C_{\to} - H(J \mid M_A).$$

By Theorem 8.2,

$$\Omega \geq 0.$$

The term Ω is information transmitted because the human does not know which part of M_H complements the AI's private state, rather than because that extra information is intrinsically needed for the decision.

Example 8.4 (Index problem)**.** Let the human know independent fair bits

$$M_H = (X_1, \ldots, X_n),$$

let the AI know an index

$$K \in \{1, \ldots, n\},$$

and let

$$J = X_K.$$

Then

$$H(J \mid K) = 1.$$

A one-way zero-error human message must permit recovery of every coordinate, so in the fixed-length zero-error setting it must distinguish all 2^n possible human strings. Thus the human sends n bits while the irreducible decision deficit is one bit, giving

$$\Omega = n - 1.$$

The coordination penalty can therefore grow without bound relative to the fundamental deficit.

8.5 Interactive querying

Allow the AI to send a query

$$Q = q(M_A)$$

before the human responds according to

$$P(Y_H \mid M_H, Q).$$

If AI-to-human communication is treated as free for the purpose of measuring human response burden, the AI may choose $Q = M_A$.

Theorem 8.5 (Interactive exact complementarity)**.** *If*

$$H(J \mid M_A, M_H) = 0,$$

and unrestricted AI-to-human querying is free, then the minimum exact human-to-AI response information is

$$C_{\leftrightarrow}^{H \to A} = H(J \mid M_A).$$

Proof. The lower bound is Theorem 8.2. For achievability, let the query reveal M_A to the human. Joint sufficiency allows the human to compute J from (M_A, M_H) and return $Y_H = J$. Then

$$I(M_H; J \mid M_A) = H(J \mid M_A) - H(J \mid M_A, M_H) = H(J \mid M_A).$$

□

Theorem 8.5 shows that, under its deliberately idealized free-query assumption, query design can eliminate coordination overhead measured only in human-to-AI response information without improving the AI's predictive model. Fewer human response bits do not necessarily imply better human modeling; they may reflect better localization of the residual decision boundary.

Theorem 8.6 (Exact bounded-query frontier for the zero-error index problem)**.** *Consider the index problem with independent fair bits $X_1, \ldots, X_n$, AI index $K \in [n]$, and target $J = X_K$. Let the deterministic AI query be $Q = q(K)$ with query alphabet size at most m, and let the human return a deterministic fixed-length response from an alphabet of size at most 2^b. Require zero-error recovery for every source string and every index. Then the minimum worst-case response length is*

$$\boxed{b^*(n, m) = \left\lceil \frac{n}{m} \right\rceil.}$$

If the query channel has a fixed-length budget of r bits, so $m \le 2^r$, then

$$\boxed{b^*(n, r) = \left\lceil \frac{n}{2^r} \right\rceil.}$$

Proof. For each query value u, define the index fiber

$$S_u = \{k \in [n] : q(k) = u\}.$$

Fix u and write $s = |S_u|$. If two source strings x^n and $\widetilde{x}^n$ differ on their restrictions to S_u, then there exists $k \in S_u$ with $x_k \neq \widetilde{x}_k$. If the human encoder returned the same message for both strings under query u, then the decoder, given the same (k, u, y), would have to output both x_k and $\widetilde{x}_k$, contradicting zero-error recovery. Hence all 2^s patterns on S_u require distinct human responses, so $b \ge s$. Therefore

$$b \ge \max_u |S_u|.$$

The fibers partition n indices into at most m cells, so the pigeonhole principle gives

$$\max_u |S_u| \ge \left\lceil \frac{n}{m} \right\rceil.$$

For achievability, partition $[n]$ into at most m cells whose sizes differ by at most one. The AI sends the cell containing K; the human returns all source bits indexed by that cell; and the AI extracts X_K. The largest cell has size $\lceil n/m \rceil$, proving equality. Substituting $m = 2^r$ gives the fixed-length query-budget form. □

Because $H(J \mid K) = 1$, the corresponding zero-error fixed-length response overhead is

$$\Omega_r^{\mathrm{fl}} = \left\lceil \frac{n}{2^r} \right\rceil - 1.$$

Thus every additional query bit approximately halves the worst-case response burden until the one-bit decision deficit is reached. The v0.7 enumeration-verified $n = 4$, $r = 1$ construction is the special case $b^* = 2$.

8.6 Lossy bounded-query frontier

The same family admits a direct connection to rate–distortion when a small average decision error is allowed. Assume for the cleanest statement that $m = 2^r$ divides n and that the query partitions the indices into equal fibers of size

$$s = \frac{n}{2^r}.$$

Let $\widehat{J}$ be the reconstructed decision and require

$$\Pr(\widehat{J} \neq J) \leq \varepsilon, \qquad 0 \leq \varepsilon \leq \frac{1}{2}.$$

Theorem 8.7 (Finite lossy bounded-query converse)**.** *In the balanced-fiber index problem above, every protocol whose human response alphabet has size at most 2^b satisfies*

$$\boxed{b \geq \frac{n}{2^r}\big[1 - h_2(\varepsilon)\big].}$$

Proof. Fix a query value u and its fiber S_u of size s. Let $\varepsilon_{i,u}$ be the conditional reconstruction error for coordinate $i \in S_u$. Since the source bits in the fiber are independent and fair,

$$H(X_{S_u} \mid Q = u) = s.$$

Binary Fano gives

$$H(X_i \mid Y, Q = u) \leq h_2(\varepsilon_{i,u}).$$

Conditional subadditivity therefore yields

$$H(X_{S_u} \mid Y, Q = u) \leq \sum_{i \in S_u} h_2(\varepsilon_{i,u}).$$

Hence

$$I(X_{S_u}; Y \mid Q = u) \geq s - \sum_{i \in S_u} h_2(\varepsilon_{i,u}).$$

Let $\overline{\varepsilon}_u = s^{-1} \sum_{i \in S_u} \varepsilon_{i,u}$. By concavity of h_2,

$$I(X_{S_u}; Y \mid Q = u) \geq s\big[1 - h_2(\overline{\varepsilon}_u)\big].$$

Averaging over the equal fibers and applying concavity once more gives

$$H(Y \mid Q) \geq s\big[1 - h_2(\varepsilon)\big].$$

Since $H(Y \mid Q) \leq b$, the claimed bound follows. □

Corollary 8.8 (Asymptotic balanced-fiber frontier)**.** *For a sequence of balanced-fiber index problems with $s = n2^{-r} \to \infty$, the converse is asymptotically tight:*

$$\boxed{b^*(n, r, \varepsilon) = n2^{-r}\big[1 - h_2(\varepsilon)\big] + o(n2^{-r}).}$$

Proof. Conditioned on a query value, the human sees an s-bit IID Bernoulli-1/2 block and the AI will later request a uniformly random coordinate from that block. Apply an asymptotically optimal binary Hamming rate–distortion code to reconstruct the fiber at average distortion at most ε. The Bernoulli-1/2 rate–distortion function is

$$R(\varepsilon) = 1 - h_2(\varepsilon).$$

The human therefore needs

$$s[1 - h_2(\varepsilon)] + o(s)$$

bits, and the AI outputs the reconstructed coordinate indexed by K. Uniformity of K within the fiber makes the decision error equal to the average per-coordinate Hamming distortion. Theorem 8.7 supplies the matching converse. □

This theorem/corollary pair separates the ideal decision-information term

$$1 - h_2(\varepsilon)$$

from the residual coordination factor

$$n2^{-r}.$$

In this symmetric family only, the asymptotic response burden factorizes as

$$\text{response burden} \approx \text{coordination factor} \times \text{decision rate–distortion}.$$

We do not claim this multiplicative factorization for general human–AI communication problems. The fractional asymptotic rates are block-coding rates, not literal fractional-bit single interactions.

Remark 8.9 (Cognitive conversion). The achievability argument in Theorem 8.5 asks the human to compute J from (M_A, M_H). Interactive querying can therefore convert coordination burden from communicated response bits into AI-to-human query complexity and human-side computation or comprehension. For bounded humans, the relevant engineering objective is at least three-way: query cost, response cost, and human cognitive cost. A full characterization is open.

8.7 Approximate complementarity

For error tolerance ε, define

$$C_{\to}(\varepsilon) = \inf I(M_H; Y_H \mid M_A)$$

subject to

$$\Pr(\widehat{J} \neq J) \leq \varepsilon.$$

Fano's inequality gives

$$C_{\to}(\varepsilon) \geq H(J \mid M_A) - h_2(\varepsilon) - \varepsilon \log_2(|\mathcal{J}| - 1).$$

Define the best joint Bayes error

$$\varepsilon^*_{AH} = \inf_{\delta} \Pr\big(\delta(M_A, M_H) \neq J\big).$$

No human-only communication protocol can achieve $\varepsilon < \varepsilon^*_{AH}$.

8.8 Decision-relevant human information ceiling

For any one-way message generated only from M_H,

$$I(Z^{\text{ref}}; Y_H \mid M_A) \leq I(Z^{\text{ref}}; M_H \mid M_A).$$

This remains a valid reference-information ceiling, but Theorem 3.3 permits a sharper decision-relative statement.

Let T_C be any deterministic decision profile through which d_C factors, and define

$$V_C^H(D) = R_{T_C}(D \mid M_A) - R_{T_C}(D \mid M_A, M_H).$$

Theorem 8.10 (Decision-relevant human side-information bound)**.** *For every feasible distortion level D,*

$$\boxed{0 \leq V_C^H(D) \leq I(T_C; M_H \mid M_A).}$$

Consequently,

$$V_C^H(D) \leq I(T_C; M_H \mid M_A) \leq I(Z^{\text{ref}}; M_H \mid M_A).$$

Proof. Nonnegativity follows because the richer decoder may ignore M_H. For the upper bound, take any feasible richer-information channel $P(A \mid T_C, M_A, M_H)$ and marginalize M_H conditional on (T_C, M_A):

$$\overline{P}(a \mid t, m_A) = \sum_{m_H} P(a \mid t, m_A, m_H) P(m_H \mid t, m_A).$$

This preserves the joint marginal law of (T_C, M_A, A) and therefore preserves expected distortion. Hence the AI-only rate is bounded by the marginalized information cost,

$$R_{T_C}(D \mid M_A) \le I(T_C; A \mid M_A).$$

For the original richer construction,

$$I(T_C; A \mid M_A) \le I(T_C; A, M_H \mid M_A)$$

and the chain rule gives

$$I(T_C; A, M_H \mid M_A) = I(T_C; M_H \mid M_A) + I(T_C; A \mid M_A, M_H).$$

Taking the infimum over richer feasible channels yields

$$R_{T_C}(D \mid M_A) \le I(T_C; M_H \mid M_A) + R_{T_C}(D \mid M_A, M_H),$$

which proves the upper bound. The final inequality follows from conditional data processing because $T_C = t_C(Z^{\mathrm{ref}})$. □

At $D = 0$ with unique optimal actions, the required action J_C is itself sufficient for exact zero-regret identification, so

$$V_C^H(0) = H(J_C \mid M_A) - H(J_C \mid M_A, M_H) = \boxed{I(J_C; M_H \mid M_A)}.$$

The full regret profile T_C can contain more information than J_C because states sharing the same optimum may assign different losses to suboptimal actions. Thus the exact zero-distortion identity should use J_C, while the general lossy bound uses the full decision profile or its minimal quotient.

9 Capability-indexed human burden

For unique exact decisions and ideal interactive querying, define the minimum human response burden

$$B_C^{\mathrm{int}} = H(J_C \mid M_C).$$

For one-way communication define

$$B_C^{\rightarrow} = B_C^{\mathrm{int}} + \Omega_C.$$

If $J_1 = f(J_2)$ and $M_1 = g(M_2)$, then

$$B_2^{\mathrm{int}} - B_1^{\mathrm{int}} = F_{1\to2}^H - L_{1\to2}^H,$$

where

$$F_{1\to2}^H = H(J_2 \mid J_1, M_2)$$

and

$$L_{1\to2}^H = H(J_1 \mid M_1) - H(J_1 \mid M_2).$$

For one-way communication,

$$B_2^{\rightarrow} - B_1^{\rightarrow} = F_{1\to2}^H - L_{1\to2}^H + (\Omega_2 - \Omega_1).$$

This yields three distinct capability effects: new human-dependent decision distinctions can raise burden; better internal models can lower the fundamental deficit; and better querying/coordination can lower Ω_C even when the deficit is unchanged.

9.1 Existence of hump-shaped grounding demand

A non-monotonic burden curve is immediate. At low capability C_1, suppose

$$H(J_1 \mid M_1) = 0.$$

At intermediate capability C_2, personalized actions become relevant and

$$H(J_2 \mid M_2) > 0.$$

At higher capability C_3, the AI internalizes those distinctions and

$$H(J_3 \mid M_3) = 0.$$

Then

$$B^{\text{int}}_{C_1} = 0, \qquad B^{\text{int}}_{C_2} > 0, \qquad B^{\text{int}}_{C_3} = 0.$$

Therefore no universal monotonic law from capability to explicit human burden is possible.

10 Constructive reference formation

The preceding sections treat human information largely as pre-existing side information. Many human-relative decisions are not of that form. A query can participate in forming the future reference.

10.1 Human transition model

Let

$$H_t \in \mathcal{H}$$

be the current human/reference state, let

$$X_t \in \mathcal{X}$$

be relevant world state, let

$$Q_t \in \mathcal{Q}_C$$

be an AI query or presentation, and let

$$\eta_{t+1} \in \mathcal{E}$$

be a human-side innovation. Define

$$F_H : \mathcal{H} \times \mathcal{X} \times \mathcal{Q}_C \times \mathcal{E} \to \mathcal{H}$$

with

$$H^{\text{ref}}_{t+1} = F_H(H_t, X_t, Q_t, \eta_{t+1}).$$

A response is generated through

$$P(Y_{t+1} \mid H^{\text{ref}}_{t+1}, Q_t),$$

and a capability-dependent decision variable may be

$$J_{C,t+1} = j_C(H^{\text{ref}}_{t+1}, X_{t+1}).$$

The edge

$$Q_t \to H^{\text{ref}}_{t+1}$$

separates constructive interaction from pure measurement.

10.2 Epistemic versus constructive uncertainty

Define total post-query decision uncertainty before the future human state is observed:

$$K^{\text{tot}}_{C,t} = H(J_{C,t+1} \mid M_t, Q_t).$$

Define the current-state epistemic contribution

$$E_{C,t} = I(J_{C,t+1}; H_t \mid M_t, Q_t),$$

and the future constructive innovation contribution

$$N_{C,t} = H(J_{C,t+1} \mid M_t, Q_t, H_t).$$

Then the conditional entropy identity gives

$$K_{C,t}^{\text{tot}} = E_{C,t} + N_{C,t}.$$

This is an exact decomposition. The term $E_{C,t}$ measures uncertainty removable by knowing the current human state better. The term $N_{C,t}$ remains even under perfect current-state knowledge because the future reference-generating process has not yet resolved.

Corollary 10.1 (Perfect-current-human-model limit)**.** *If, under finite-alphabet regularity,*

$$H(H_t \mid M_t) \to 0,$$

then the epistemic contribution can vanish, but this alone does not imply

$$N_{C,t} \to 0.$$

Thus a perfect model of the current human is not equivalent to a perfect predictor of future legitimately formed reference decisions.

If the transition F_H is deterministic and all of its inputs are known to the AI, then $N_{C,t} = 0$. Therefore constructive persistence requires actual residual innovation rather than an appeal to deliberation as such.

10.3 Lossy constructive grounding

Define

$$R_{C,t}^{\text{con}}(D) = \inf I(H_{t+1}^{\text{ref}}; A_{t+1} \mid M_t, Q_t, H_t),$$

subject to

$$\mathbb{E}[d_C(H_{t+1}^{\text{ref}}, A_{t+1})] \le D.$$

For unique exact decisions,

$$R_{C,t}^{\text{con}}(0) = N_{C,t}.$$

10.4 Constructive persistence condition

If there exists $r > 0$ such that, in the protected allowed process,

$$\liminf_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} R_{C,t}^{\text{con}}(D) \ge r,$$

then at least r fresh decision-relevant bits per unit time must enter after perfect current-state knowledge, unless another permitted channel observes the same innovations before action.

This is a conditional persistence theorem, not a universal fact about humans.

11 Integrity and endogenous targets

11.1 Allowed causal policy class

Let

$$\Pi^{\text{allow}}$$

be the externally specified class of permitted interaction/control policies. For

$$\pi \in \Pi^{\text{allow}},$$

write

$$Z_t^{\text{ref},\pi}$$

for the protected reference process induced under π.

The information objective is evaluated only over policies in Π^{allow}. Integrity is therefore a constraint on the admissible causal process, not another term that can be traded freely against information gain.

11.2 Why unconstrained endogenous grounding collapses

Proposition 11.1 (Unrestricted endogenous-reference collapse)**.** *If the AI is permitted to choose a policy π for which*

$$Z_{t+1}^{\mathrm{ref},\pi} = g(M_t)$$

deterministically, then

$$R_{C,t+1}^{\pi}(0) = 0.$$

This is not evidence that the AI learned the protected reference; it may simply have made the target predictable. The proposition shows why any endogenous-reference theory without a protected policy class is degenerate.

11.3 Learning, legitimate constitution, and steering

A human response may reflect three causally distinct mechanisms:

- epistemic revelation of a pre-existing reference component;
- legitimate constitution of a reference through allowed deliberation;
- disallowed steering that changes the target to make it easier to predict or optimize.

These components are not generally additive random variables. They are causal categories. Information theory alone cannot classify them from final agreement statistics.

Proposition 11.2 (Observational non-identifiability)**.** *If two causal models generate the same observational law over the measured variables (Q, Y, A) but differ in the latent causal pathway through the human/reference state, then no statistic measurable only from samples of (Q, Y, A) can distinguish the two models.*

Thus integrity requires protected causal variables, interventions, path-specific assumptions, or an exogenously chosen reference class.

11.4 Conservative reference refinement

As a tractable toy notion of legitimate deliberation, let

$$\mathcal{O}_H(h) \subseteq \mathcal{A}_C$$

be the currently acceptable-action set. A deliberation map

$$T_Q : \mathcal{H} \to \mathcal{H}$$

is a conservative reference refinement if

$$\mathcal{O}_H(T_Q(h)) \subseteq \mathcal{O}_H(h).$$

It resolves ambiguity without introducing an action that was outside the prior acceptable set. This is intentionally stronger than a general theory of human value formation and should not be treated as a universal normative criterion.

Definition 11.3 (Integrity-compatible closure)**.** Constructive grounding closes with integrity if $N_{C,t} \to 0$ along trajectories induced by some policy $\pi \in \Pi^{\mathrm{allow}}$ while the required task distortion remains satisfied.

Proposition 10.1 shows that $N_{C,t}$ can be forced toward zero by unrestricted target manipulation, whereas the perfect-current-human-model limit in Section 9 does not by itself imply $N_{C,t} \to 0$. The live question is therefore whether constructive grounding can close *inside* the allowed policy class. Legitimate deliberation may reduce the constructive rate, leave it positive, or reorganize the decision problem; rate reduction by itself does not certify legitimate closure.

12 Preference compression and asymptotic closure

Constructive grounding does not imply permanent human interaction. Repeated deliberation can compress into a stable higher-level representation.

Let

$$\Theta \in \mathcal{T}$$

be a stable preference statistic. Suppose future decisions factor as

$$J_{C,t} = f_C(X_t, \Theta).$$

If the AI eventually learns Θ exactly and observes X_t, then

$$H(J_{C,t} \mid M_{C,t}) \to 0.$$

Thus a finite learnable sufficient preference representation is a strong closure mechanism.

Conversely, after conditioning on the best stable learned representation Θ^* and all permitted non-human observations $X_{1:T}$, define the residual reference-decision entropy rate

$$\overline{H}_C^{\text{ref}} = \liminf_{T\to\infty} \frac{1}{T} H(J_{C,1:T} \mid \Theta^*, X_{1:T}).$$

For lossy tasks, the corresponding object should be a conditional sequential rate–distortion rate rather than entropy rate.

The asymptotic research question becomes

$$\lim_{C\to\infty} \overline{R}_C^{\text{ref}}(D) \quad ?$$

with three basic regimes:

$$\overline{R}_C^{\text{ref}}(D) \to 0$$

for informational closure;

$$\overline{R}_C^{\text{ref}}(D) \to r > 0$$

for persistent grounding; and growth of

$$\overline{R}_C^{\text{ref}}(D)$$

for an expanding decision-information frontier.

13 Protocol burden, information rate, and consequence

The master theorem gives a fundamental lower envelope. Human interaction burden can be larger because of protocol constraints. In an exact one-way setting,

$$B_C^{\rightarrow} = H(J_C \mid M_C) + \Omega_C.$$

Interactive querying can reduce Ω_C without changing $H(J_C \mid M_C)$.

Separately, let

$$\mathcal{L}_C(D)$$

be an outcome-loss functional mapping residual distortion to consequence. Two systems can have the same $R_{\text{dec}}(C, M_C, D)$ but radically different downstream harm. Therefore the information race and harm race are distinct.

The earlier binary toy model makes this separation explicit. With residual error p_C, effective dimension k_C, amplification a_C, and tolerated outcome loss D, the model-specific grounding law is

$$G_C(D) = k_C \left[h_2(p_C) - h_2\left(\min\left\{ p_C, \frac{D}{a_C} \right\} \right) \right].$$

If

$$p_C \asymp C^{-\alpha}, \qquad k_C \asymp C^{\gamma}, \qquad a_C \asymp C^{s},$$

then one can have

$$\gamma < \alpha$$

while

$$s > \alpha.$$

In that regime, the amount of missing information can vanish while ungrounded loss diverges. This is the "vanishing bits, diverging value" existence result. It remains model-specific and is not the master theorem.

14 Geometry and lower bounds

Conditional decision rate–distortion is the central object. Packing, Fano, and covering arguments are tools for bounding it.

Suppose $z_1, \ldots, z_N$ form a $2r$-packing under a relevant metric and let J index the true packed state. If a reconstruction has expected distortion at most $D < r$, nearest-neighbor decoding has error probability at most D/r. Fano's inequality therefore yields a lower bound of the form

$$G(D) \geq H(J \mid M) - h_2(D/r) - \frac{D}{r} \log_2(N-1).$$

For a skewed prior, the conditional entropy $H(J \mid M)$ rather than raw $\log_2 N$ is the correct average-case complexity term.

Upper bounds follow from explicit quantizers or decision partitions. If a quantizer $Q = q(Z, M)$ yields a reconstruction satisfying distortion D, then

$$G(D) \leq H(Q \mid M).$$

Matching lower and upper bounds, rather than a lower bound alone, are required to establish an asymptotic scaling law.

15 Empirical status inherited from v0.4

The existing empirical miniature does not estimate $R_{\mathrm{dec}}(C, M, D)$ directly. It tests whether an observable consensus-error proxy survives stronger internal information.

15.1 Internal diversity gap

At ensemble size $E = 5$ and 2,000 training examples per member, the measured all-member-wrong rates were:

Condition	All items	Annotator-unanimous
Bootstrap same-family	0.049	0.031
Disjoint same-family	0.033	0.016
Diverse shared-data	0.022	0.009
Diverse disjoint-data	0.016	0.005

Moving from bootstrap same-family to diverse disjoint reduced consensus-wrong error by roughly two-thirds overall and by 84% on annotator-unanimous items. The surviving residual increasingly concentrated on annotator-contested items, reaching approximately 79% in the most-diverse condition. The 0.5% clean residual is therefore only an upper-bound proxy for information-limited error and had not visibly plateaued in model diversity.

15.2 Query-strategy race after diversification

The final balanced errors in the diverse three-family experiment were:

Strategy	Final balanced error
Uncertainty sampling	0.0855
Failure-memory	0.1052
Oracle all majority-wrong	0.0892
Oracle clean majority-wrong	0.0863

Once internal diversity was strengthened, disagreement-driven querying matched or beat targeted shared-failure localization. The corrected empirical order is therefore: exhaust internal diversity first; then measure whether a clean systematic residual survives; only then interpret external labels as evidence about information-limited grounding demand.

16 Direct experimental measurements

The master theorem suggests measuring the grounding curve directly rather than relying only on consensus proxies.

For an external-information budget B, define the best achievable distortion

$$D_C(B).$$

The inverse-type quantity estimates

$$G_C(D).$$

A controlled experiment can restrict external answers to known message capacities, such as a binary answer, one of four choices, or one of eight choices. This yields empirical points on $D_C(B)$ while holding internal compute fixed.

The v0.6 theory suggests measuring separately:

1. internal exhaustion: residual error after maximal permitted internal diversity, aggregation, retrieval, and verification;
2. decision refinement: whether higher capability creates additional protected distinctions that change optimal action;
3. grounding response curves $D_C(B)$;
4. human unique decision value $V_C^H(D)$;
5. coordination overhead Ω_C under one-way versus interactive protocols;
6. dynamic residual rate $R_{C,t}(D)$ after conditioning on the complete causal past;
7. constructive innovation $R_{C,t}^{\mathrm{con}}(D)$ after conditioning additionally on the current human state;
8. harm amplification $\mathcal{L}_C(D)$;
9. integrity variables needed to distinguish learning, legitimate constitution, and steering.

16.1 First direct grounding-response measurement and a protocol-density hypothesis

Executing the controlled-channel program on the v0.4 task at fixed internal compute, a diverse three-family ensemble trained on a 2,000-example base was paired with three explicit external reply alphabets. Four retraining rounds were run to approximately matched cumulative raw reply-capacity budgets near 380 bits, with two seeds.

At the final budget, a one-bit label channel targeted by ensemble disagreement reached balanced error 0.082; the same one-bit label channel with random targeting reached 0.104; and a localization channel that presented eight unlabeled items and accepted one of nine replies—one named incorrect item or "none"—used approximately

$$\log_2 9 \approx 3.17$$

raw bits per interaction and reached balanced error 0.099.

The registered prediction that the richer localization channel would dominate per external bit was falsified. The observed failure is consistent with a *protocol-density hypothesis*: localization or group-testing-style alphabets become more attractive when the addressed error/event density is sparse enough that a negative reply certifies many candidates, whereas dense queried regions can waste the extra localization alphabet. In the tested disagreement region the measured error density was approximately 0.4, so an eight-item shortlist contained at least one error with probability near 0.98 under an independence heuristic. In that regime, the "none" branch almost never paid while each successful localization still spent roughly 3.17 raw reply bits to identify one item.

This is an empirical design principle, not a proved law. The data establish only that, on this task and among these three protocols, the richer localization alphabet was less bit-efficient than per-item uncertainty-targeted labels. Whether localization dominates when the queried residual becomes sufficiently sparse is a registered follow-up hypothesis. More generally, channel value depends jointly on query capacity, response capacity, the geometry of the residual, and the cost assigned to human computation.

17 HG-v2 implementation update and architectural implications

The first HG-v2 implementation cycle predates several of the v0.9 theorems but provides useful evidence about the architecture-level decomposition. The results below are empirical engineering results, not

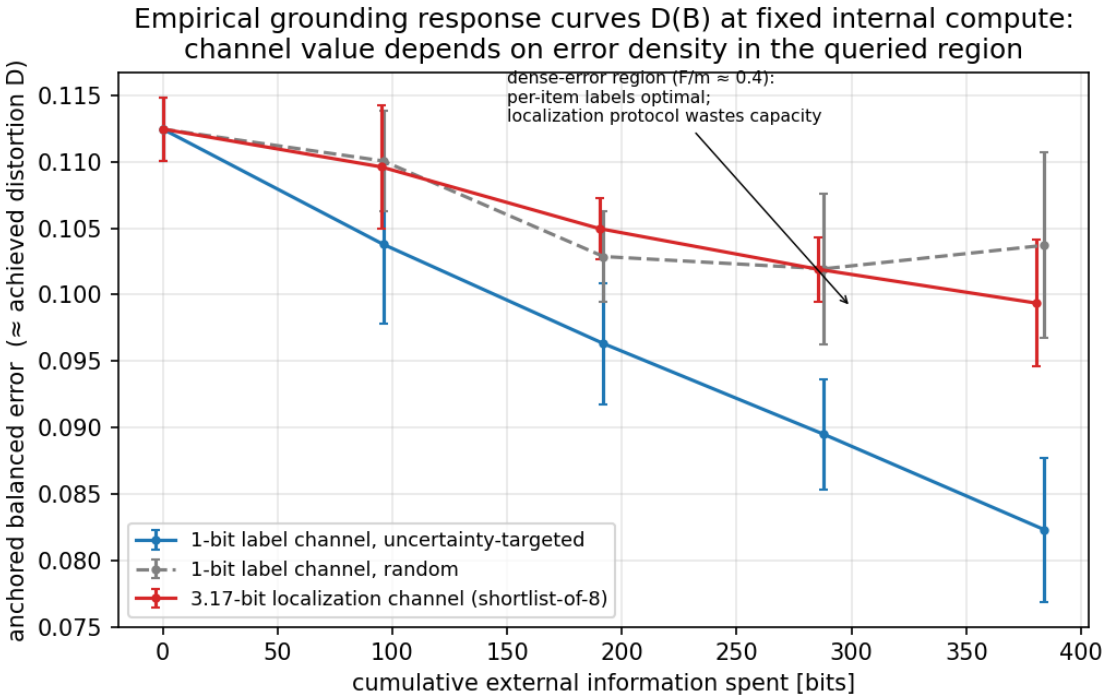


Figure 1: First empirical grounding-response curves $D_C(B)$ at fixed internal compute. Error bars show the supplied two-seed variation. The uncertainty-targeted one-bit label channel outperformed both random labels and the tested shortlist-of-eight localization protocol at the matched final raw reply-capacity budget.

theorem tests beyond the deliberately synthetic sanity checks.

17.1 G0–G2 results

The index sanity experiment used 48,000 episodes over

$$n \in \{2, 4, 8, 16, 32, 64\}.$$

ACT remained near chance, generic asking used exactly n response bits, targeted asking used exactly one bit, and both sufficient ask policies achieved perfect decisions. A negative-control source that lacked X_K paid one well-formed response bit but remained near chance. This directly demonstrates that communication spent and decision information acquired are different quantities.

In the noisy-index routing experiment, a logistic router trained on internal-agreement features beat both always-ACT and always-ASK at seven of eight tested information prices. The sweep was

$$\lambda \in \{0.01, 0.02, 0.05, 0.10, 0.15, 0.20, 0.30, 0.40\}.$$

The router tied always-ACT only at the largest price. This establishes routing value on the controlled task but does not by itself establish calibrated estimation: threshold policies can agree even when probability estimates differ materially.

The first real-LM query-compression run used a frozen Qwen2.5-3B-Instruct model on 1,000 route-choice items. The three grounded arms had nearly equal final accuracy, while human-response costs were

$$B_{\text{generic}} = 5.00, \qquad B_{\text{oracle}} = 1.00, \qquad B_{\text{synth}} = 2.52.$$

Thus the synthesized query saved 2.48 bits relative to generic escalation at essentially matched final distortion. The normalized oracle-gap closure was

$$\eta_Q = \frac{5.00 - 2.52}{5.00 - 1.00} \approx 0.62.$$

The result is limited to one task family, one model, and one seed; the model's fully grounded rule-following ceiling remained only about 0.77.

17.2 Routing, soundness, and minimality are distinct

The query synthesizer named a decision-relevant variable on every evaluated item, but only about 58% of queries were exactly minimal. Its query-size distribution was strongly bimodal: approximately 58% requested one variable, about 34% escalated to all five variables, and the remainder lay between. Thus mean response cost alone conceals two qualitatively different modes.

This motivates three distinct architectural quantities:

- **Routing:** whether fresh information should be acquired at all;
- **Soundness:** whether the query requests information sufficient to resolve the required decision;
- **Minimality:** conditional on soundness, how much excess response capacity the query requests above the decision-theoretic optimum.

The frozen query synthesizer emitted no "no query needed" decisions and asked on all items, including roughly 49% whose visible state already closed the decision. Meanwhile the separate router handled the ACT/ASK distinction well on its synthetic task. The current evidence therefore favors an explicit router followed by a query synthesizer rather than one undifferentiated prompting module.

17.3 Measurement corrections

The implementation cycle exposed four measurement issues. First, coordination overhead is meaningless for insufficient protocols: zero communication by ACT can produce a formally negative subtraction $C - H(J \mid M_A)$ without resolving the deficit, while an irrelevant one-bit message can falsely appear optimal. Accordingly, $\Omega(D)$ should be defined only over protocols satisfying the target distortion. Second, query soundness and minimality must be reported separately. Third, aggregate accuracy can hide label bias and must be split by correct-action class. Fourth, an existing chat-template rank-scoring path used a space-prefixed label continuation where the correct tokenization required the bare label; prior rank-scored chat-template experiments using that path must be rechecked before they are treated as evidence.

17.4 Training and model-architecture implications

The theoretical quotients above suggest a concrete training principle: do not train the internal representation to reconstruct the complete world when the objective is decision grounding. Instead learn a state S_t that approximately preserves both

$$d(z, a)$$

and

$$P(S_{t+1} \mid z, a).$$

A representation objective can therefore contain a decision-profile term and a controlled-transition term,

$$\mathcal{L}_{\text{repr}} = \lambda_d \mathcal{L}_{\text{decision}} + \lambda_p \mathcal{L}_{\text{transition}},$$

with empirical estimates of $(\varepsilon_d, \varepsilon_p)$ tied to Theorem 7.8. This is a more targeted objective than generic world reconstruction and provides a route toward certifiable horizon-dependent representation quality.

For HG-v2, the current preferred architecture is:

internal decision state → grounding router → query synthesizer → external source → final decision.

The router should estimate whether the current residual decision requirement is worth external acquisition, rather than merely model confidence. The query synthesizer should optimize soundness first and response minimality second. In the bounded-query toy family, the correct target is the query–response frontier, not textual brevity.

The near-term training order should remain staged rather than end-to-end. First freeze a capable base LM and validate the capability floor. Second train or calibrate the decision-state estimator/router using counterfactual ACT/ASK outcomes. Third train or constrain the query synthesizer using structured query specifications, explicit response budgets, and soundness/minimality objectives. Fourth introduce a learned dynamic decision quotient whose auxiliary losses preserve decision profiles and controlled transitions. Only after these components pass independent gates should LoRA or joint training be attempted.

Several specific improvements are now motivated:

- add a **capability-floor probe** before any grounding-efficiency comparison, using D_C^{blind}, D_C^{full}, and when available U_C;
- train the router and query synthesizer as **separate modules**, since the first experiment found strong targeting with essentially no ability to abstain from querying;
- optimize query **minimality under a soundness constraint**, including hard variable caps, explicit response-bit budgets, or candidate-query reranking;
- retain a **grounding memory** in the causal state so information acquired once can reduce future conditional decision entropy rather than being repeatedly reacquired;
- evaluate learned state representations through $(\varepsilon_d, \varepsilon_p)$ and horizon-sensitive decision degradation, not reconstruction loss alone;
- keep **human response cost**, **AI query cost**, and provisional proxies for human cognitive burden separate rather than combining them prematurely into one scalar;
- preserve internal diversity and disagreement routing, but treat failure memory as an update to M_t rather than the central architectural mechanism;
- delay joint optimization until the router, query synthesizer, representation, and grounded-decision decoder each pass their own sufficiency tests.

The architecture implication is therefore not simply "ask better questions." It is to learn a compact dynamically decision-relevant state, estimate the residual value of grounding, issue the smallest sound query for the unresolved decision quotient, retain reusable grounded information, and apply the answer with a decoder that has already passed the capability floor.

18 What the theory proves and does not prove

Claim	Status
Ideal one-shot grounding equals conditional decision rate–distortion.	Proved under unrestricted ideal encoder assumptions.
Nested extra internal information cannot increase ideal grounding demand.	Proved.
Decision-refining capability cannot reduce grounding at fixed internal information.	Proved under the explicit projection condition.
Action-set enlargement alone cannot increase grounding monotonically.	False; a universally acceptable added action gives an explicit counterexample.
The information-race decomposition equals capability pressure minus internal-learning relief.	Exact algebraic identity.
Unique zero-regret grounding equals $H(J_C \mid M_C)$.	Proved.
IID fresh decision innovations require $R_C(D)$ bits per step in the ideal block model.	Proved.
The master local lower bound applies to arbitrary dependent causal processes.	Proved locally.
A complete global feedback equality is given by the local sum.	Not proved; generally false without additional causal structure; this does not weaken the local inequality in Theorem 5.1.
One-way human response cost is at least $H(J \mid M_A)$ in the jointly sufficient exact case.	Proved.
Free unrestricted querying can reduce ideal human response cost to $H(J \mid M_A)$.	Proved under free, unrestricted query capacity and free human computation.
In the deterministic fixed-length zero-error index family, bounded query and response budgets satisfy $b^*(n, r) = \lceil n/2^r \rceil$.	Proved exactly.
In balanced lossy index families, $b^*(n, r, \varepsilon) = n2^{-r}[1 - h_2(\varepsilon)] + o(n2^{-r})$.	Finite converse plus asymptotically tight achievability proved.
Interactive querying eliminates total coordination cost.	False as stated; it can convert response bits into query complexity and human computation.
Coordination overhead Ω can dominate the fundamental deficit.	Demonstrated and quantitatively characterized in the index family.

Decision rate–distortion depends only on a sufficient decision profile when distortion factors through it.	Proved by Theorem 3.3.
Human side-information value is bounded by $I(T_C; M_H \mid M_A)$.	Proved; sharper than the full-reference bound.
Exogenous causal ideal grounding equals nonanticipative decision rate–distortion.	Proved under exogenous causal-source and unrestricted ideal-message assumptions.
Causal decision rate–distortion depends only on the causal decision-profile process.	Proved under stage-local distortion and exogenous dynamics.
Exact dynamic decision quotients preserve the controlled causal information–distortion frontier.	Proved for finite controlled Markov systems under distortion preservation and controlled lumpability.
Approximate dynamic decision quotients preserve the frontier up to explicit distortion slack Δ_T^{abs} and information penalty η_T.	Proved for finite action alphabets; same-D closeness is not claimed.
A model below its capability floor has high grounding demand.	False interpretation; below D_C^{full} the target is infeasible for the restricted decoder, so more message bits alone cannot solve it.
Perfect modeling of the current human removes future constructive innovation.	False in general; only true when the relevant dynamics become predictable.
Human-specific grounding remains positive as $C \to \infty$.	Not proved; requires a positive residual protected human/reference rate.
Real AI capability generally satisfies decision refinement.	Structural hypothesis, not a theorem.
Consensus-wrong error is information-limited error.	Not established; existing experiments show substantial proxy inflation.

19 Counterexamples that the framework must survive

- If the protected environment is deterministic and perfectly observed, then $R_{\mathrm{dec}}(C, M, D)$ can be zero.
- If a stronger capability introduces a universally acceptable action, grounding can decrease despite a larger action set.
- If human-private information never changes an acceptable action, then $I(Z^{\mathrm{ref}}; M_H \mid M_A) > 0$ can coexist with $V_C^H(D) = 0$.
- If multiple optimal action sets overlap, the entropy of an arbitrary tie-broken action label can overstate grounding.
- If an AI steers the observed human target toward its own prediction, unanchored grounding can collapse without learning the protected reference.
- If all future human decisions factor through a stable finite statistic Θ that the AI learns completely, ongoing explicit human grounding can vanish.

20 Unified interpretation

The theory can now be represented as a stack.

First, an allowed policy $\pi \in \Pi^{\mathrm{allow}}$ induces a protected reference process

$$Z_t^{\mathrm{ref},\pi}.$$

Second, the AI's complete pre-grounding information is M_t. Third, the residual ideal decision-information requirement is

$$R_{C,t}^{\pi}(D).$$

Fourth, an actual external channel must supply at least that much reference-relevant information locally, while any excess is protocol or acquisition inefficiency. Fifth, the outcome-loss functional $\mathcal{L}_C(D)$ determines the value of obtaining the information. Integrity is enforced by the restriction $\pi \in \Pi^{\mathrm{allow}}$ rather than traded off as an ordinary information term.

This leads to the compact backbone

$$\text{fresh external information} \;\geq\; \text{residual decision rate–distortion.}$$

In exogenous causal systems the corresponding block quantity is a nonanticipative rate–distortion functional of the causal decision-profile process. In controlled systems the relevant state can be reduced further to a dynamic decision quotient preserving both current distortion and future quotient transitions. Approximate learned quotients trade representation compression against the explicit horizon-dependent penalties Δ_T^{abs} and η_T. Human-specific grounding is the part of the residual that can only be observed or legitimately constituted through the protected human/reference process. Coordination overhead is the gap between that fundamental lower bound and constrained human communication. Constructive grounding changes the future source distribution but not the local data-processing lower bound. Capability scaling changes both the decision problem and the internal side information.

21 Next theoretical steps

Version v0.9 resolves much of the exogenous sequential theory and the exact/approximate finite controlled quotient problem. The remaining mathematical priorities are narrower:

1. extend the controlled information–distortion theory to explicit causal side information, observation channels, and partially observed systems rather than fully observed quotient states;
2. derive stationary or discounted infinite-horizon versions of Theorem 7.8, including conditions under which the finite-horizon continuity penalty has a finite asymptotic rate;
3. formalize constructive human/reference dynamics as a controlled source under Π^{allow} and identify integrity constraints that rule out entropy reduction by disallowed target steering;
4. connect the non-unique zero-error decision problem formally to graph/hypergraph entropy and functional source coding with decoder side information;
5. extend bounded-query results from the symmetric index family to general functional side-information problems, stochastic queries, variable-length codes, and explicit human-computation costs;
6. empirically test whether learned dynamic decision quotients with small $(\varepsilon_d, \varepsilon_p)$ predict downstream grounding efficiency and long-horizon decision degradation;
7. replicate the HG-v2 routing/query result across capable model scales and a second controlled task family after rechecking the affected chat-template rank-scoring path.

22 Conclusion

Version v0.9 retains the framework's single informational backbone while extending it from one-shot decisions to exogenous causal processes and finite controlled Markov systems. Exact dynamic decision quotients preserve the complete causal information–distortion frontier, while approximate quotients preserve it up to explicit distortion and directed-information continuity penalties. The implementation results add a complementary engineering lesson: useful grounding requires not only missing information, but also a capable decoder, a router that decides whether acquisition is needed, and a query protocol that is both sound and minimal. For a protected reference process and a fixed capability-dependent decision problem, every actual external grounding channel must locally supply at least the residual conditional decision rate–distortion left after the AI's complete internal state and query context are known.

Static grounding, IID innovation, dependent open-system persistence, human–AI complementarity, coordination overhead, and constructive preference formation are not separate theories. They arise by changing the source process, the side information, or the allowed communication protocol around the same data-processing lower bound.

The framework does not prove that humans remain necessary forever. It identifies the condition that would make persistent human-originating grounding necessary: after all learnable stable structure, all permitted non-human observations, and optimal internal processing are conditioned out, the protected reference process must continue to generate positive decision-relevant rate–distortion. If that rate vanishes, informational closure is possible. If it stays positive, persistent grounding is required. If capability causes the protected decision-information rate to grow faster than internal learning resolves it, an expanding frontier is possible.

The asymptotic question is therefore no longer "does smarter AI need humans?" It is

$$\lim_{C\to\infty} \overline{R}_C^{\mathrm{ref}}(D) \quad ?$$

with protocol overhead Ω_C, consequence $\mathcal{L}_C(D)$, and integrity $\pi \in \Pi^{\text{allow}}$ measured separately.

Within the binary model-specific family, the sharpest surviving possibility remains a regime of vanishing information but diverging value: when the earlier power-law parameters satisfy $\gamma < \alpha$ while consequence amplification satisfies $s > \alpha$, the missing information can tend to zero even as the cost of omitting it diverges. Nothing in the master theorem forbids this combination. It is therefore more informative to measure the grounding curve and the consequence map separately than to assume either permanent supervision or harmless informational closure.

A Symbol and function-signature dictionary

Object	Type / signature	Meaning
C	capability index	Indexes a decision/action capability and associated internal system.
Z_C^{ref}	$\Omega \to \mathcal{Z}_C$	Protected reference-relevant state.
M_C	$\Omega \to \mathcal{M}_C$	Complete usable AI information before new grounding.
$\mathcal{A}_C$	action/policy set	Actions available at capability C.
u_C	$\mathcal{A}_C \times \mathcal{Z}_C \to \mathbb{R}$	Reference-relative utility.
d_C	$\mathcal{Z}_C \times \mathcal{A}_C \to [0, \infty)$	Decision/reference distortion.
D	$[0, \infty)$	Target expected distortion.
Y	$\Omega \to \mathcal{Y}$	Fresh external message.
R_{dec}	$(C, M, D) \mapsto$ bits	Ideal conditional decision rate–distortion.
J_C	$\Omega \to \mathcal{A}_C$ or decision alphabet	Unique correct decision when defined.
ϕ	$\mathcal{A}_{C_2} \to \mathcal{A}_{C_1}$	Action projection witnessing decision refinement.
$F_{1\to 2}$	$D \mapsto$ bits	Capability pressure at fixed old internal information.
$L_{1\to 2}$	$D \mapsto$ bits	Internal-learning relief at fixed new capability.
Q_t	$q_t(M_t)$	AI query/intervention before fresh observation.
$R_{C,t}^{\pi}(D)$	$D \mapsto$ bits	Local protected decision rate–distortion under policy π.
M_A	$\Omega \to \mathcal{M}_A$	AI private information in the two-agent model.
M_H	$\Omega \to \mathcal{M}_H$	Human private information.
$C_{\to}$	scalar in bits	Minimum exact one-way human-to-AI response information.
Ω	scalar in bits	Coordination overhead above the fundamental decision deficit.
H_t	$\Omega \to \mathcal{H}$	Current human/reference state.
F_H	$\mathcal{H} \times \mathcal{X} \times \mathcal{Q}_C \times \mathcal{E} \to \mathcal{H}$	Human/reference transition under query and innovation.
$E_{C,t}$	scalar in bits	Current-state epistemic contribution.
$N_{C,t}$	scalar in bits	Future constructive innovation contribution.
$R_{C,t}^{\text{con}}(D)$	$D \mapsto$ bits	Constructive rate–distortion after current human state is known.
Π^{allow}	set of policies	Externally specified allowed causal interaction policies.
Θ	stable latent/statistic	Learned higher-level preference representation when one exists.
$\mathcal{L}_C(D)$	loss functional	Consequence of operating at residual distortion D.
$R_Z^{\text{na},T}(D)$	causal rate–distortion functional	Nonanticipative decision-information requirement for an exogenous process.
$S_t = \phi(Z_t)$	quotient state	Dynamic decision quotient preserving distortion and quotient transitions.
ε_d	nonnegative scalar	Maximum within-class distortion-model error of an approximate quotient.

ε_p	$[0,1]$	Maximum total-variation transition error of an approximate quotient.
Δ_T^{abs}	nonnegative scalar	Finite-horizon distortion slack induced by approximate quotienting.
η_T	nonnegative scalar in bits	Directed-information continuity penalty for a finite action alphabet.
D_C^{full}	distortion level	Best distortion attainable by capability class Π_C with full available grounding information.
U_C	$[0,1]$	Fraction of externally available decision value exploited by the restricted decoder.

B Proof dependency map

Result	**Load-bearing ingredients**
Theorem 2.3	Conditional data-processing inequality; unrestricted ideal encoder; choose $Y = A$ for converse.
Theorem 3.1	Nested internal information; ability to ignore extra information; chain rule / conditional data processing.
Theorem 3.5	Nonnegative regret; unique optimum; entropy identity.
Theorem 4.2	Pointwise distortion-preserving action projection; conditional data processing.
Theorem 5.1	Local causal action condition; conditional data processing; definition of local decision rate–distortion.
Theorem 6.1	IID source; additive distortion; mutual-information decomposition; convexity of rate–distortion.
Theorem 8.1	Complete human side information must suffice; converse by sending M_H.
Theorem 8.2	Exact recovery; human message generated from M_H; conditional data processing.
Theorem 8.5	Joint sufficiency; free unrestricted AI query; return J after revealing M_A; response-information accounting only.
Theorem 3.3	Distortion factorization through a deterministic decision profile; channel lifting and Markovization; conditional data processing.
Theorem 8.6	Deterministic fixed-length zero-error index family; query fibers; injectivity on each fiber; pigeonhole principle; balanced partition achievability.
Theorem 8.7 / Corollary 8.8	Binary Fano, conditional subadditivity, entropy concavity, and Bernoulli Hamming rate–distortion achievability.
Theorem 8.10	Decision-profile reduction; channel marginalization; conditional chain rule; conditional data processing.
Constructive decomposition	Conditional entropy identity $H(J \mid M, Q) = I(J; H \mid M, Q) + H(J \mid M, Q, H)$.

C References and neighboring machinery

The information-theoretic primitives used here are standard or closely related to existing literatures. The project-specific contribution under investigation is the capability-indexed synthesis of decision refinement, internal-information relief, dynamic reference innovation, constrained human communication, constructive reference formation, harm amplification, and integrity.

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External-review status. The theorem family has now undergone the finite assumption-knockout audit in Appendix D; no counterexample to any theorem as stated was found, while each tested load-bearing assumption has an explicit failure witness when removed. This is not a substitute for independent external mathematical review. The new sequential and controlled claims should still be checked carefully against the established nonanticipative rate–distortion, directed-information, bisimulation, and approximate state-abstraction literatures. The bounded interactive frontier remains solved only for the symmetric index family; its general functional and cognitive-cost versions remain open. The conditional-entropy continuity constants used in Theorem 7.8 are deliberately conservative and should receive independent mathematical review.

D Assumption-knockout audit

Following the v0.6 hardening program, individual assumptions were removed and finite counterexamples were sought. Every tested load-bearing assumption below has an explicit failure witness. Two finite constructions—the non-unique-optimum example and the bounded-query index example—were checked by exhaustive enumeration in the accompanying review work. No counterexample to any theorem as stated was found.

K1. Ideal encoder side information in Theorem 2.3

If the encoder sees only Z^{ref} and not M, the ideal equality can fail by an unbounded factor. In the index family, $R_{\mathrm{dec}} = H(J \mid M) = 1$ bit, while a fixed-length zero-error one-way source ignorant of the decoder's index must distinguish all 2^n source strings and can require n response bits. The unrestricted encoder-side-information assumption is therefore load-bearing.

K2. Functional nesting in Theorem 3.1

Neither larger entropy nor larger aggregate mutual information with Z^{ref} can replace decision-relevant nesting. Let $Z = (Z_a, Z_b)$ with independent uniform components, let $J = Z_a$, let $M_1 = Z_a$, and choose M_2 to contain independent information only about Z_b, with enough entropy that $I(Z; M_2) > I(Z; M_1)$. Then $R_{\mathrm{dec}}(M_1, 0) = 0$ while $R_{\mathrm{dec}}(M_2, 0) = 1$. What matters is the Blackwell/garbling-type relation, not the scalar amount of information.

K3. Uniqueness in Theorem 3.5

Let Z be uniform on $\{0,1\}$ with actions $\{a,b,c\}$ and zero-regret sets

$$\mathcal{O}(0) = \{a,c\}, \qquad \mathcal{O}(1) = \{b,c\}.$$

The constant rule $A = c$ has zero regret and $I(Z;A) = 0$, while a tie-breaking rule selecting a at 0 and b at 1 has one bit of entropy. Thus the formula $H(J \mid M)$ requires uniqueness; the set-valued functional-coding formulation is essential otherwise.

K4. Zero distortion in Theorem 3.5

For a uniform binary exact-decision variable under Hamming distortion and any $0 < D < 1/2$,

$$R(D) = 1 - h_2(D) < 1 = H(J).$$

Thus equality with full decision entropy is special to the zero-distortion exact-decision case.

K5. Pointwise projection in Theorem 4.2

Action-set enlargement alone does not imply monotonic grounding. If C_2 adds an action a^* satisfying

$$d_{C_2}(z, a^*) = 0$$

for every z, then the stronger action set admits zero grounding at zero distortion even when the old problem required positive information. The projection condition is therefore a genuine structural hypothesis.

K6. Query measurability in Theorem 5.1

If Q_t may depend directly on Z_t^{ref}, take

$$Q_t = Z_t^{\mathrm{ref}}, \qquad Y_t = \mathrm{const}, \qquad A_t = Q_t.$$

Zero distortion is possible while

$$I(Z_t^{\mathrm{ref}}; Y_t \mid M_t, Q_t) = 0.$$

The missing information entered through Q_t. Hence either Q_t must be measurable from M_t, as stated, or the query information itself must be charged.

K7. Cumulative tightness

The local sum in Theorem 5.1 is a lower bound, not a general operational equality. Information acquired at one time can enter later internal states and reduce future residual requirements, while the sequence of pointwise optimal test channels need not be jointly realizable under feedback. This does not provide a counterexample to the local inequality; it is precisely why the global problem calls for a directed-information or nonanticipative rate–distortion formulation.

K8. Bounded query capacity in Theorem 8.5

The v0.7 $n = 4$ exhaustive-enumeration witness is now subsumed by Theorem 8.6. In the deterministic fixed-length zero-error index family,

$$b^*(n,r) = \left\lceil \frac{n}{2^r} \right\rceil.$$

Thus the free-query one-bit response optimum is attained exactly when the query alphabet can localize the desired index sufficiently finely. The broader stochastic, variable-length, functional, and cognitive-cost frontiers remain open.

K9. Human computation in Theorem 8.5

The achievability proof reveals M_A to the human and asks the human to compute J from (M_A, M_H). Therefore the theorem eliminates only the measured human-to-AI response overhead under its accounting convention. With bounded human computation or comprehension, a query-capacity/response-burden/cognitive-cost frontier replaces a single scalar Ω.

Audit item	**Result**
Counterexample to any theorem as stated	None found in the finite audit.
Load-bearing assumptions witnessed by failure	Ideal encoder side information; nesting/garbling; uniqueness; $D = 0$; pointwise projection; query measurability; sufficient query capacity; free human computation.
Enumeration-verified constructions	Non-unique tie-breaking example; bounded-query $n = 4$ index example, now subsumed by Theorem 8.6.